



HTML DOM

1. Introduction to HTML DOM
2. HTML Node and Tree Structure
3. Accessing HTML elements

- It is a language and platform independent programming interface consisting of standard properties and methods.
- The HTML DOM provides a structured representation of an HTML document.
- This allows to access, modify, and interact with its elements, attributes, and content using JavaScript.
- This is essential for creating dynamic web pages and web applications.

- DOM views HTML document as different nodes arranged in the form of tree structure.
- Every part of HTML document becomes a node in the tree.
- Node types
 - Nodes in this tree structure can be element node, attribute node, text node, comment node or document node.
 - Every type of node has few important properties: nodeName, nodeValue and nodeType.
 - Text node gets created for text content inside the HTML element which is considered as child node of element node.

- Node relationships in the tree structure:
 - Only one root node
 - Only one parent for any node
 - Any no of children for any node
 - Nodes having same parent - siblings nodes
 - Nodes without child - leaf nodes

- Different ways are used to access the nodes means HTML elements in the node tree(HTML document)
 - using id
 - `getElementById()`: returns the first element with matching id
 - using tag name
 - `getElementsByTagName()`: returns collection of all nodes with specified tag name
 - using class name
 - `getElementsByClassName()`: returns collection of all nodes with specified class name

- using CSS selector
 - `querySelector()`: returns first node matching CSS selector
 - `querySelectorAll()`: returns a collection of all nodes matching CSS selector
- using name
 - `getElementsByName`: returns a collection of all nodes matching specified name attribute